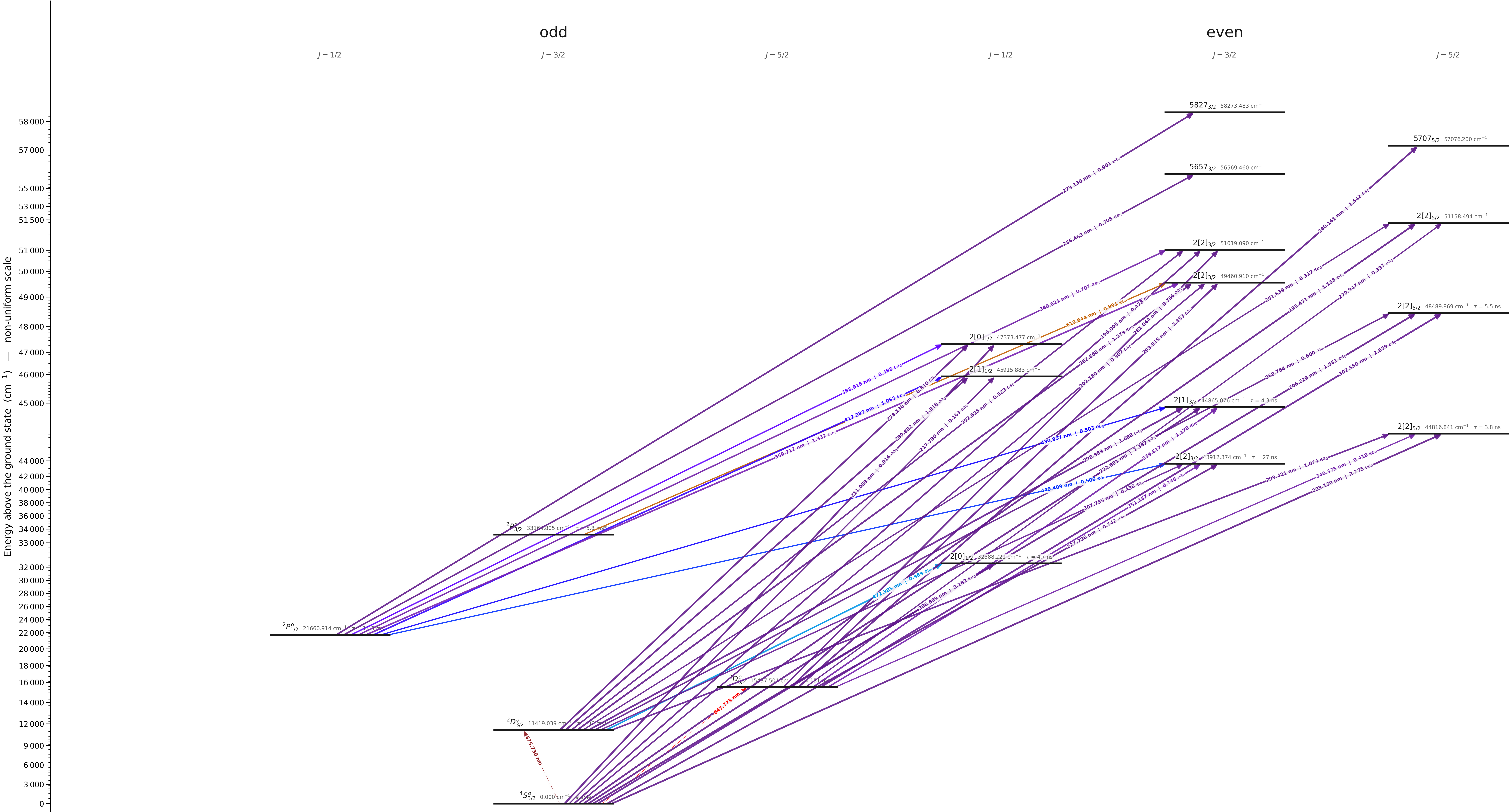
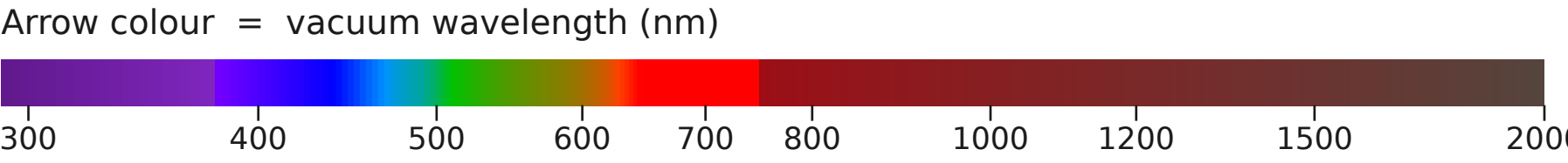


Bismuth

Bi (Bi I). Strongest classified transitions below 2 μm (NIST): 18 levels, 39 lines.

Bi⁺ ionisation limit 58761.65 cm⁻¹ = 7.28552 eV (170.179 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J || e r || J' \rangle |$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J e r J' \rangle $ ($e a_0$)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$4S^{\circ}_{3/2} - 2[0]_{1/2}$	306.8593	306.7702	976.9703	2.182	1.670e+08	33.86	78.5	resonance line for collinear and in-source laser spectrosc...
$4S^{\circ}_{3/2} - 2D^{\circ}_{5/2}$	647.7732	647.5943	462.8046	-	-	1.054e-06	-	magnetic-dipole line used for parity-nonconservation opti...
$4S^{\circ}_{3/2} - 2D^{\circ}_{3/2}$	875.7304	875.4900	342.3342	-	2.250e+01	4.421e-06	81	magnetic-dipole line of the Oxford parity-nonconservation...

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

²⁰⁹Bi hyperfine structure ($I = 9/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$4S^{\circ}_{3/2}$	-446.9420(10)	-304.6540(20)	F=3: +3547.64 F=4: +1962.97 F=5: -208.27 F=6: -3093.02
$2D^{\circ}_{3/2}$	-1231.020	-652.500	F=3: +9856.85 F=4: +5367.77 F=5: -651.39 F=6: -8472.51
$2D^{\circ}_{5/2}$	2502.860	23.000	F=2: -34403.78 F=3: -26902.10 F=4: -16897.18 F=5: -4386.71 F=6: +10632.17 F=7: +28162.92
$2P^{\circ}_{1/2}$	11260.200	-	F=4: -30965.55 F=5: +25335.45
$2[0]_{1/2}$	4920.8(6)	-	F=4: -13532.20 F=5: +11071.80
$2P^{\circ}_{3/2}$	491.028	978.640	F=3: -3602.44 F=4: -2290.75 F=5: -39.50 F=6: +3559.10

Measured A, B constants; sources in the data file.